python.py - C:/Users/ASUS/python.py (3.11.1)

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```
import heapq

GOAL = (1, 2, 3, 4, 5, 6, 7, 8, 0)
MOVES = {"Up": -3, "Down": 3, "Left": -1, "Right": 1}

def is_solvable(state):
    tiles = [t for t in state if t != 0]
    inv = sum(1 for i in range(len(tiles)) for j in range(i + 1, len(tiles)) if tiles[i] > tiles[j])
    return inv % 2 == 0

def manhattan(state):
    total = 0
    for idx, tile in enumerate(state):
        if tile:
            goal_idx = tile - 1
            total += abs(idx // 3 - goal_idx // 3) + abs(idx % 3 - goal_idx % 3)
    return total

def neighbours(state):
    b = state.index(0)
    for name, d in MOVES.items():
        if name == "Left" and b % 3 == 0:
            continue
        if name == "Right" and b % 3 == 2:
            continue
        n = b + d
        if 0 <= n < 9:
            s = list(state)
            s[b], s[n] = s[n], s[b]
            yield name, tuple(s)

def solve(start):
    if not is_solvable(start):
        return None
    counter = 0
    open_heap = [(manhattan(start), 0, counter, start)]
    parent = {start: (None, None)}
    best_g = {start: 0}
    while open_heap:
        f, g, _, state = heapq.heappop(open_heap)
        if state == GOAL:
            path = []
            while parent[state][0] is not None:
                prev, move = parent[state]
                path.append((move, state))
                state = prev
            path.reverse()
```

Ln: 79 Col: 0

27°C Light rain

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IDLE Shell 3.11.1

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```
Python 3.11.1 (tags/v3.11.1:a7a450f, Dec  6 2022, 19:58:39) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:/Users/ASUS/python.py ==========
Start:
1 3 6
5 _ 2
4 7 8

Step 1: move blank Right
1 3 6
5 2 _
4 7 8

Step 2: move blank Up
1 3 _
5 2 6
4 7 8

Step 3: move blank Left
1 _ 3
5 2 6
4 7 8

Step 4: move blank Down
1 2 3
5 _ 6
4 7 8

Step 5: move blank Left
1 2 3
_ 5 6
4 7 8

Step 6: move blank Down
1 2 3
4 5 6
_ 7 8

Step 7: move blank Right
1 2 3
4 5 6
7 _ 8

Step 8: move blank Right
1 2 3
4 5 6
```

Ln: 51 Col: 0

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